

ORKHON, Mongolia

WMO: 442420

Lat: 49.1436N

Lon: 105.4022E

Elev: 2181

StdP: 77.72

Time Zone: 8.00 (E08)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-37.1	-35.0	-40.3	0.1	-36.9	-38.2	0.1	-34.9	8.6	-12.7	7.1	-15.2	0.5	110	0.604

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.4	31.9	16.5	29.8	16.2	27.9	15.9	19.8	26.8	18.6	25.3	17.7	24.3	3.0	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.7	16.6	23.7	16.5	15.4	22.2	15.5	14.4	21.2	67.3	26.8	62.9	25.1	59.3	24.4	23.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.3	8.0	6.7	DB	-39.1	35.5	2.8	2.3	-41.2	37.2	-42.8	38.5	-44.4	39.8	-46.4	41.5	(4)
(5)			WB	-39.2	21.1	2.7	1.4	-41.2	22.1	-42.8	22.9	-44.3	23.7	-46.3	24.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	0.2	-25.2	-20.2	-6.2	4.7	11.2	17.8	20.5	17.7	11.3	2.1	-10.8	-21.6	(6)	
	DBStd	16.68	5.36	6.73	7.24	4.99	4.92	4.08	3.10	3.33	4.42	4.74	7.20	6.19	(7)	
	HDD10.0	4545	1092	846	501	171	45	1	0	1	38	246	625	981	(8)	
	HDD18.3	6774	1350	1079	759	411	224	57	12	52	214	502	875	1239	(9)	
	CDD10.0	971	0	0	0	11	83	234	327	239	76	2	0	0	(10)	
	CDD18.3	160	0	0	0	0	5	41	81	32	2	0	0	0	(11)	
	CDH23.3	2041	0	0	0	8	163	591	834	376	68	1	0	0	(12)	
	CDH26.7	689	0	0	0	1	47	216	304	111	11	0	0	0	(13)	
(14) Wind	WSAvg	1.9	0.9	1.1	2.3	3.0	3.1	2.6	2.0	1.9	2.1	1.9	1.3	0.9	(14)	
(15) Precipitation	PrecAvg	300	2	2	5	7	22	56	80	77	31	10	5	5	(15)	
	PrecMax	434	6	6	16	17	65	142	132	187	59	44	15	13	(16)	
	PrecMin	208	0	0	1	2	2	25	28	35	7	1	1	1	(17)	
	PrecStd	55	2	2	4	4	15	27	32	36	16	10	3	3	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-7.2	-0.1	15.2	24.3	31.4	34.8	35.1	32.6	28.4	20.4	9.7	-3.3	(19)	
		MCWB	-8.6	-3.4	5.1	9.6	14.3	15.9	18.6	17.0	14.6	8.9	3.3	-5.9	(20)	
	2%	DB	-11.0	-3.9	10.9	20.2	27.7	31.4	32.2	29.6	25.2	16.3	5.6	-7.3	(21)	
		MCWB	-12.1	-6.1	3.2	7.5	12.4	15.3	17.8	17.0	13.7	7.3	0.3	-9.0	(22)	
	5%	DB	-14.2	-6.9	7.6	17.3	24.8	29.1	30.0	27.3	22.7	13.4	2.1	-10.3	(23)	
		MCWB	-15.1	-8.5	1.4	6.5	11.5	15.0	17.5	16.6	12.6	5.6	-1.4	-11.4	(24)	
	10%	DB	-16.9	-9.9	4.7	14.3	21.9	26.7	27.8	25.1	20.0	10.7	-0.7	-12.8	(25)	
		MCWB	-17.6	-11.2	-0.4	5.2	10.2	14.5	17.1	16.2	11.5	4.2	-3.1	-13.6	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-8.5	-3.5	5.9	10.3	15.9	19.1	21.6	20.6	16.2	9.3	3.3	-5.6	(27)	
		MCDB	-7.3	-0.1	13.9	22.0	27.2	28.7	30.4	27.5	24.3	19.0	9.5	-3.2	(28)	
	2%	WB	-12.0	-6.0	3.5	8.5	13.6	17.7	20.2	19.1	14.7	7.6	0.8	-9.0	(29)	
		MCDB	-11.0	-3.9	10.4	18.4	25.2	26.3	27.7	25.9	22.8	15.8	5.1	-7.5	(30)	
	5%	WB	-15.1	-8.4	1.6	7.1	12.0	16.8	19.3	17.8	13.4	5.8	-1.3	-11.4	(31)	
		MCDB	-14.3	-6.9	7.2	16.4	23.1	24.4	25.8	24.2	21.2	12.2	1.8	-10.3	(32)	
	10%	WB	-17.6	-11.1	-0.3	5.6	10.7	15.8	18.3	17.0	12.1	4.4	-3.1	-13.6	(33)	
		MCDB	-17.0	-10.0	4.4	13.7	20.9	23.9	24.7	22.7	19.1	10.2	-0.5	-12.8	(34)	
(35) Mean Daily Temperature Range	MDBR		9.9	13.1	14.1	14.3	15.1	14.0	12.4	12.5	14.5	14.0	11.4	9.4	(35)	
	5% DB	MCDBR	12.4	14.9	16.7	20.1	21.0	18.3	16.3	17.3	19.3	18.5	14.6	11.9	(36)	
		MCWBR	11.4	12.8	9.8	10.2	9.3	6.9	5.6	6.6	8.6	10.2	10.0	10.5	(37)	
	5% WB	MCDBR	12.4	14.7	16.2	18.6	18.8	13.8	12.8	13.4	16.7	17.4	13.7	11.9	(38)	
		MCWBR	11.5	12.7	9.8	9.9	8.9	5.9	5.6	6.1	7.7	9.8	9.6	10.4	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.204	0.219	0.250	0.282	0.308	0.318	0.311	0.295	0.261	0.240	0.220	0.206	(40)	
	taud		2.406	2.402	2.443	2.450	2.398	2.449	2.506	2.538	2.593	2.584	2.483	2.398	(41)	
	Ebn at Noon		912	968	981	972	952	941	943	948	960	935	881	857	(42)	
	Edh at Noon		60	78	91	102	112	108	100	93	78	65	56	53	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.76	2.91	4.35	5.41	6.32	6.48	6.07	5.21	4.37	2.91	1.80	1.36	(44)	
	RadStd		0.07	0.13	0.17	0.25	0.37	0.39	0.35	0.34	0.23	0.15	0.09	0.08	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (7 neighbors)		N/A	N/A	-2.00	N/A	N/A	-0.59	N/A	N/A	N/A	N/A

Nomenclature: See separate page